

SYED MASRUR AHMED

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EDUCATION

Dartmouth College

Hanover, NH

Bachelor of Arts, Major in Mathematics and Computer Science, GPA 3.94/4.00

Expected June 2028

- **Relevant Coursework:** Data Structures & Algorithms, Discrete Math, Fullstack Development, Computer Architecture, Machine Learning and Statistical Methods.

SKILLS

Programming Languages: Python, C, C++, Assembly, Java, JavaScript, TypeScript, PHP, Rust, C#, SQL, PostgreSQL, HTML, CSS

Technologies: Git, Docker, Next.js, React, Flask, FastAPI, PyTorch, Pandas, Unity, Flutter, AWS (Bedrock, Lambda, S3)

WORK EXPERIENCE

DALI Lab (Dartmouth Applied Learning & Innovation)

Hanover, NH

Software Engineer (Part-time)

September 2025 – Present

- **Evergreen AI (Student Wellness Platform):** Architected a FastAPI backend integrating AWS Bedrock and the Model Context Protocol (MCP) to execute secure LLM tool calls, autonomously translating natural language queries into multi-source API data retrieval and generating dynamic visualizations on Flutter frontend.
- **SimReach (Healthcare SaaS):** Implemented an AWS-based decision support tool to optimize physician outreach, implementing a data validation algorithm for EHR uploads and building full-stack onboarding flows with automated Lambda-driven cron jobs.

Department of Mathematics, Dartmouth College

Hanover, NH

Machine Learning Research Assistant (Part-time)

March 2025 – Present

- Derived asymptotic conditions under which a single attention layer implements the left shift operator under sinusoidal positional encoding, identifying a sharp phase transition that predicts when positional retrieval collapses as sequence length and embedding dimension scale.
- Built a PyTorch training and SVD pipeline validating these predictions, showing a rank-4 key projection recovers over 99.9% of target attention across embedding dimensions 16 to 64 by learning a low-rank correction concentrated in high-frequency rotation blocks.

The National Law Review

Highland Park, IL (Remote)

Software Engineering Intern

June 2026 – August 2026

- Built end-to-end Python web scraping workflows to aggregate unstructured data from diverse municipal websites, powering a centralized public notices database proposed to the New Hampshire governing body.
- Shipped full-stack platform features using custom Drupal PHP modules and responsive frontend interfaces, automating ETL migrations while implementing robust anti-scraping system architecture.

PROJECTS

MaskProxy | [Github](#) | Rust, Cloudflare Pingora, Redis, LanceDB, Next.js

March 2026 – April 2026

- Wrote a high-performance privacy middleware in Rust using Cloudflare Pingora to intercept LLM API calls, utilizing NER models (GLiNER/DeBERTa-v3) to detect and mask PII with deterministic tokens stored in Redis for stateful rehydration.
- Designed a semantic routing engine with LanceDB and BGE-Small embeddings to autonomously direct sensitive prompts to local Phi-3.5 instances, while developing a Next.js dashboard for real-time observability into masking and routing decisions

Trippee | [trippee-ai.vercel.app](#) | [Github](#) | Next.js 15, TypeScript, Supabase, Google Gemini

December 2025 – January 2026

- Developed a real-time collaborative trip-planning platform using Next.js 15 and Supabase, utilizing WebSockets to synchronize geospatial map state, live user presence, and shared rich-text editing across client sessions.
- Integrated an agentic AI workflow using Google Gemini for autonomous database interactions via tool calling, and applied K-Means clustering algorithms to automate itinerary structuring and route optimization.

HONORS AND AWARDS

- **Academic Distinctions:** Rufus Choate Scholar (Awarded to the top 5% of students), Jack Byrne Scholar in Mathematics; Academic Citations for Excellence in Linear Algebra and Machine Learning.
- **International Achievements:** Highest Score in the World (A-levels Chemistry); Highest Score in Bangladesh (A-level Further Mathematics)